

Mengqi Shi

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Research Interests

Multimodal Learning, Computer Vision, AI systems

Education

University of Hawaii at Manoa, Honolulu, USA
Ph.D in Computer Science

Aug 2025 - Present
Advisor: Prof.Haopeng Zhang

Nanyang Technological University, Singapore, Singapore
M.S. in Computer Control & Automation
Thesis: Adversarial Cross-Modal Domain Adaptation for Multi-Modal Semantic Segmentation in Autonomous Driving

Aug 2021 – Aug 2022
Advisor: Prof.Lihua Xie

Southeast University, Nanjing, China
B.E. in Automation

Sep 2017 – Jun 2021

Research Experience

Ph.D. Researcher, Aloha Lab, University of Hawaii at Manoa Sep 2025 - Present

- Developed HAVEN, a hierarchically aligned multimodal benchmark for unified video understanding. (Preprinted)
- Designed evaluation protocols and conducted large-scale experiments on MLLMs beyond surface-level QA or summarization.

Research Assistant, IoT Lab, Nanyang Technological University Mar 2022 – Apr 2022
Supervisor: Dr. Jianfei Yang, Prof. Lihua Xie

- Conducted research on cross-modal domain adaptation for autonomous driving perception.
- Conducted research on sensor data collection, focusing on data fusion.
- Investigated adversarial learning methods for multi-modal semantic segmentation across image and LiDAR modalities.
- Collaborated with a team of 5 graduate students on multi-modal learning.
- A publication at ICARCV 2022.

Industry Experience

Sinopac Institute of Safety Engineering
AI Engineer

China
Aug 2022 - Feb 2025

- Designed ML and CV pipelines for domain-specific intelligent systems.
- Built and analyzed large-scale industry datasets for risk prediction and decision-making leveraging SQL and data-driven evaluation.
- Authored 4 patents in applied AI systems.

Publications

- Shi, Mengqi, et al. “Adversarial Cross-modal Domain Adaptation for Multi-modal Semantic Segmentation in Autonomous Driving.” *2022 17th International Conference on Control, Automation, Robotics and Vision (ICARCV). IEEE, 2022.*
- Shi, Mengqi, and Haopeng Zhang. “HAVEN: Hierarchically Aligned Multimodal Benchmark for Unified Video Understanding.” Under review arXiv:2605.19223 (2026).

Teaching Experience

Teaching Assistant , Programming Language Theories	Fall 2025
Teaching Assistant (Lab lecturer) , Introduction to Computer Science	Spring 2026
Teaching Assistant , Human-computer Interaction Systems	Spring 2026